

Energie éolienne

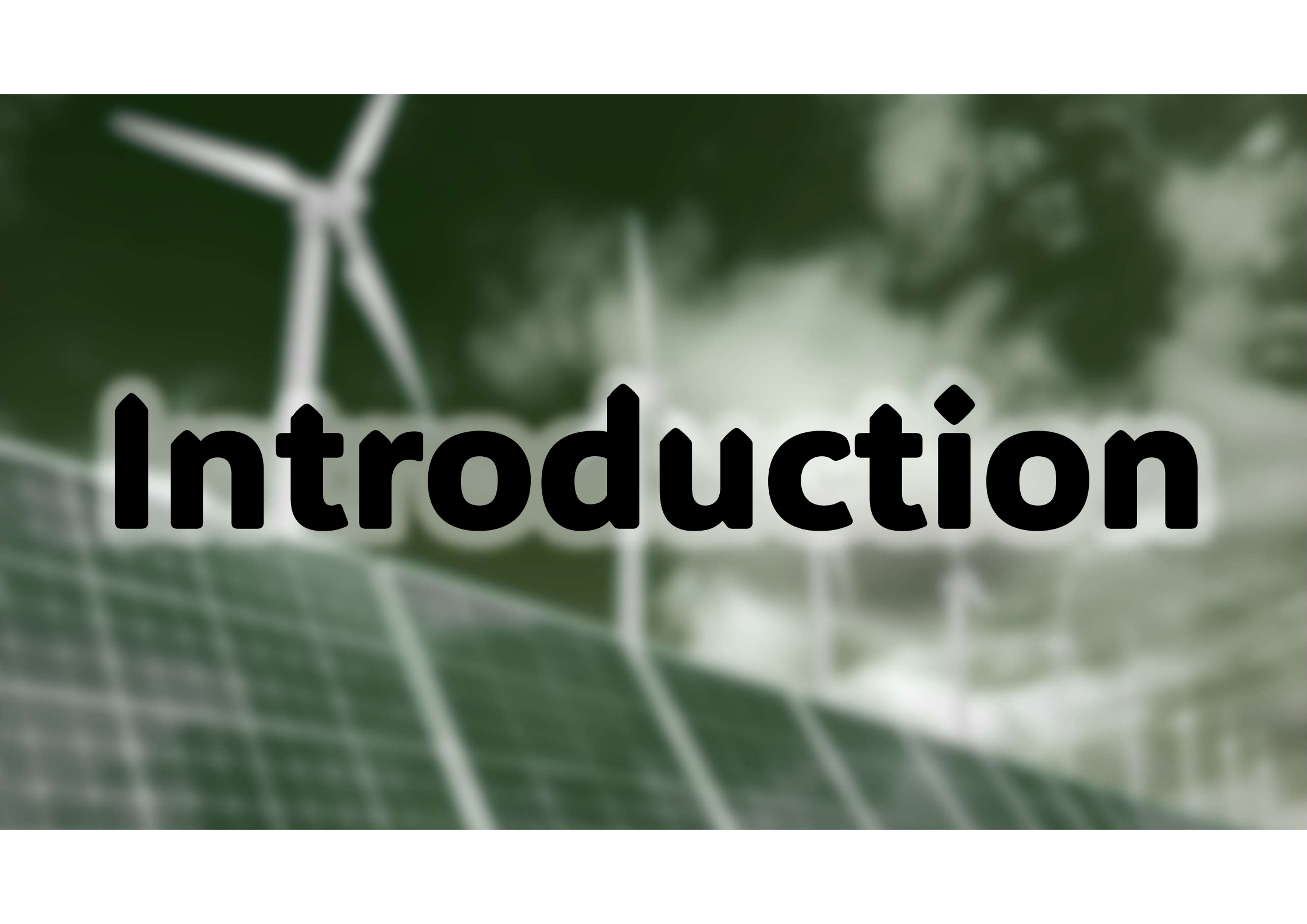
Pour SAM (System Advisor Model)

Plan



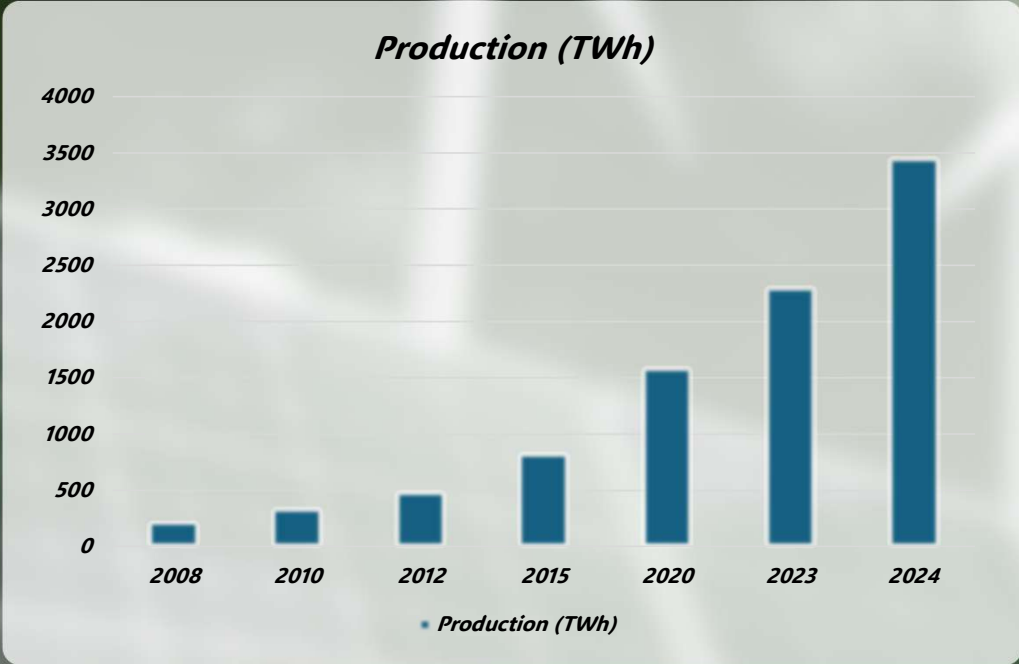
Plan

- 1. Introduction**
- 2. Origine du vent et phénomènes quotidiens**
- 3. Turbulence et cisaillement**
- 4. Principe de fonctionnement**
- 5. Loi fondamentale de la physique**
- 6. Limite de Betz**
- 7. Les données météorologiques et Sources**
- 8. Traitement des données**

The background of the slide is a blurred photograph. On the left side, a white wind turbine is visible, its blades extending upwards. The rest of the background is filled with out-of-focus green leaves and branches, suggesting a natural, outdoor setting.

Introduction

Introduction

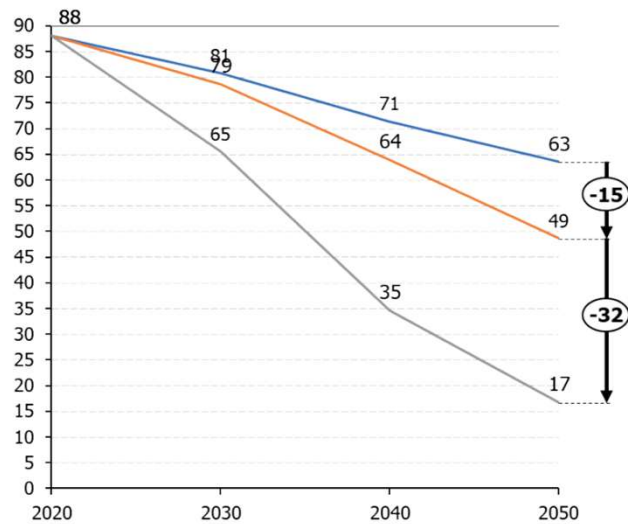


Nom du parc	Année	Puissance (MW)	Localisation	Créateur du projet
Tarfaya	2014	301	Tarfaya	Tarec (NAREVA/ENGIE)
Midelt	2020	210	Midelt	Masen
Khalladi (Melloussa)	2017	120	Melloussa	Acwa Power/TAQA
Akhfennir 1	2013	101.87	Akhfennir	NAREVA
Akhfennir 2	2016	95.2	Akhfennir	NAREVA
Essaouira	2006	60	Essaouira	ONEE
Foum El Oued (Laâyoune)	2013	50	Laâyoune	NAREVA
Haouma (Fnideq)	2013	50	Fnideq	ONEE
Al Koudia Al Baida (Fnideq)	2000	50	Fnideq	Masen/ONEE
LAFARGE (Tétouan)	2010	32	Tétouan	Lafarge Ciments Maroc
Ciments du Maroc (Laâyoune)	2011	5	Laâyoune	Ciments du Maroc
Taza	2022	87	Taza	EDF Renewables/Mitsubishi
Oualidia	2021	18	Oualidia	InnoVent
Tanger I	2011	140	Tanger	ONEE
Aftissat I (Boujdour)	2019	201.6	Boujdour	ENERGIE EOLIENNE DU MAROC (NAREVA)
Aftissat II (Boujdour)	2023	200	Boujdour	ENERGIE EOLIENNE DU MAROC (NAREVA)
Boujdour	2023	300	Boujdour	NAREVA

Introduction

Taux de dépendance énergétique, %

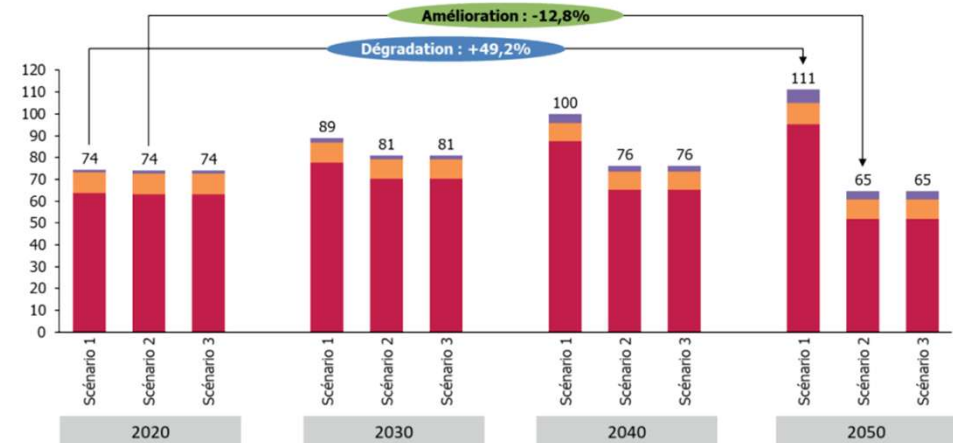
— Scénario I — Scénario II : accéléré — Scénario III : sur orbite



Source : Modélisation du CESE

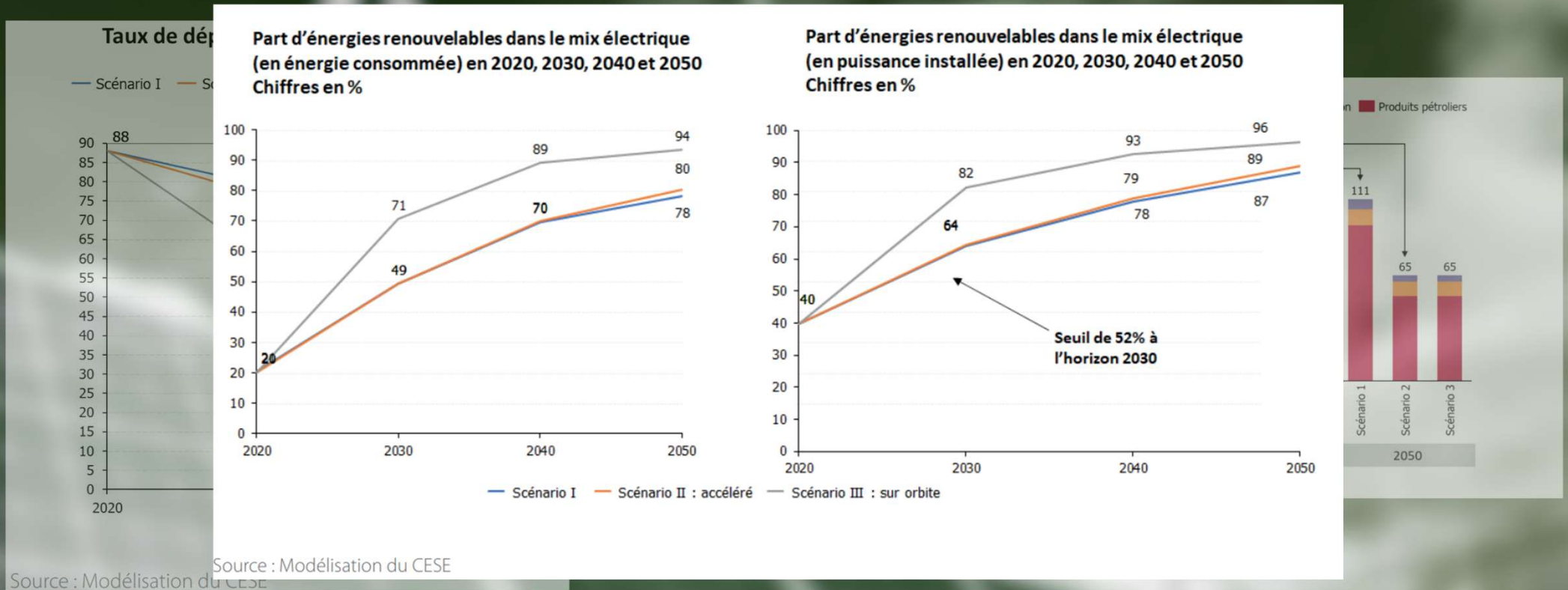
Impact sur la facture énergétique en 2020, 2030, 2040 et 2050 Chiffres en milliards de DH

■ Gaz naturel ■ Charbon ■ Produits pétroliers



Source : Modélisation du CESE

Introduction



Introduction

2 370
MW

35%

Du mix électrique marocain en 2025



Introduction

**2 370
MW**

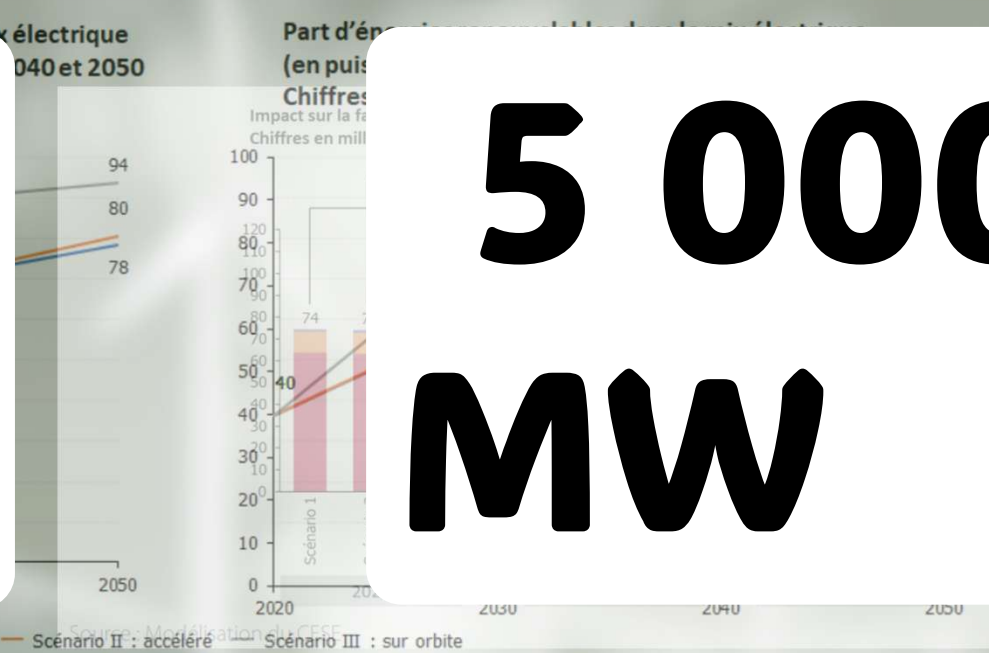
35%

Du mix électrique marocain en 2025

**5 000
MW**

40%

**Du mix électrique (de source
renouvelables) marocain en 2030**



The background of the slide is a blurred photograph. On the left side, a white wind turbine is visible, its blades extending towards the top left. The rest of the background is filled with out-of-focus green foliage, likely trees or bushes, creating a natural, outdoor setting.

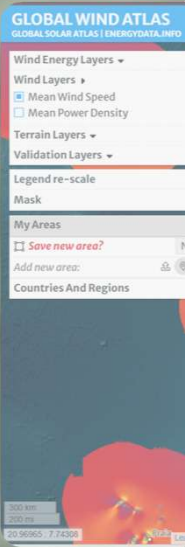
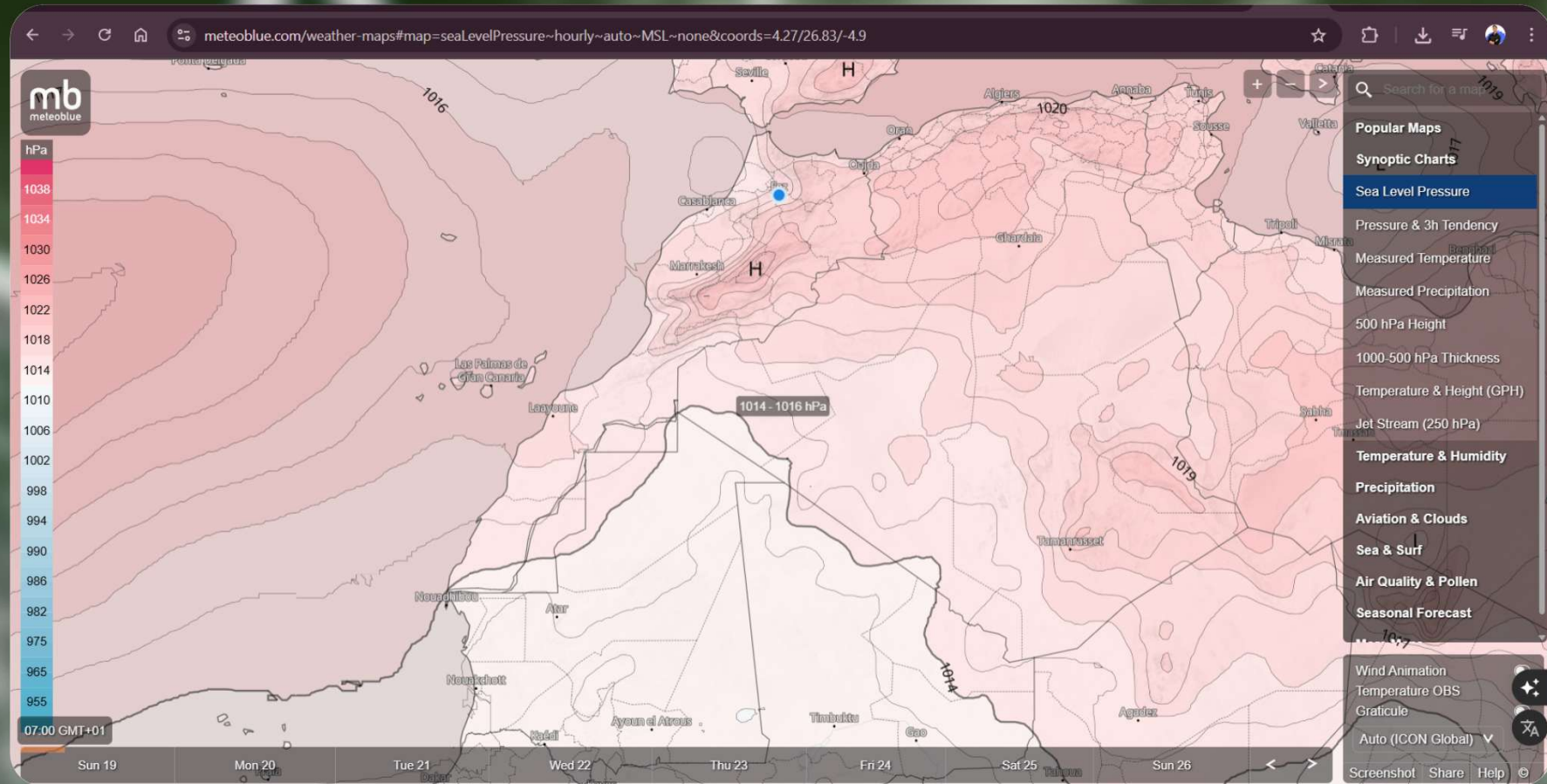
Origine du vent et phénomènes quotidiens

Origine du vent et phénomènes quotidiens

chauffage solaire inégal

Pressure Gradient

Wind (The Result)

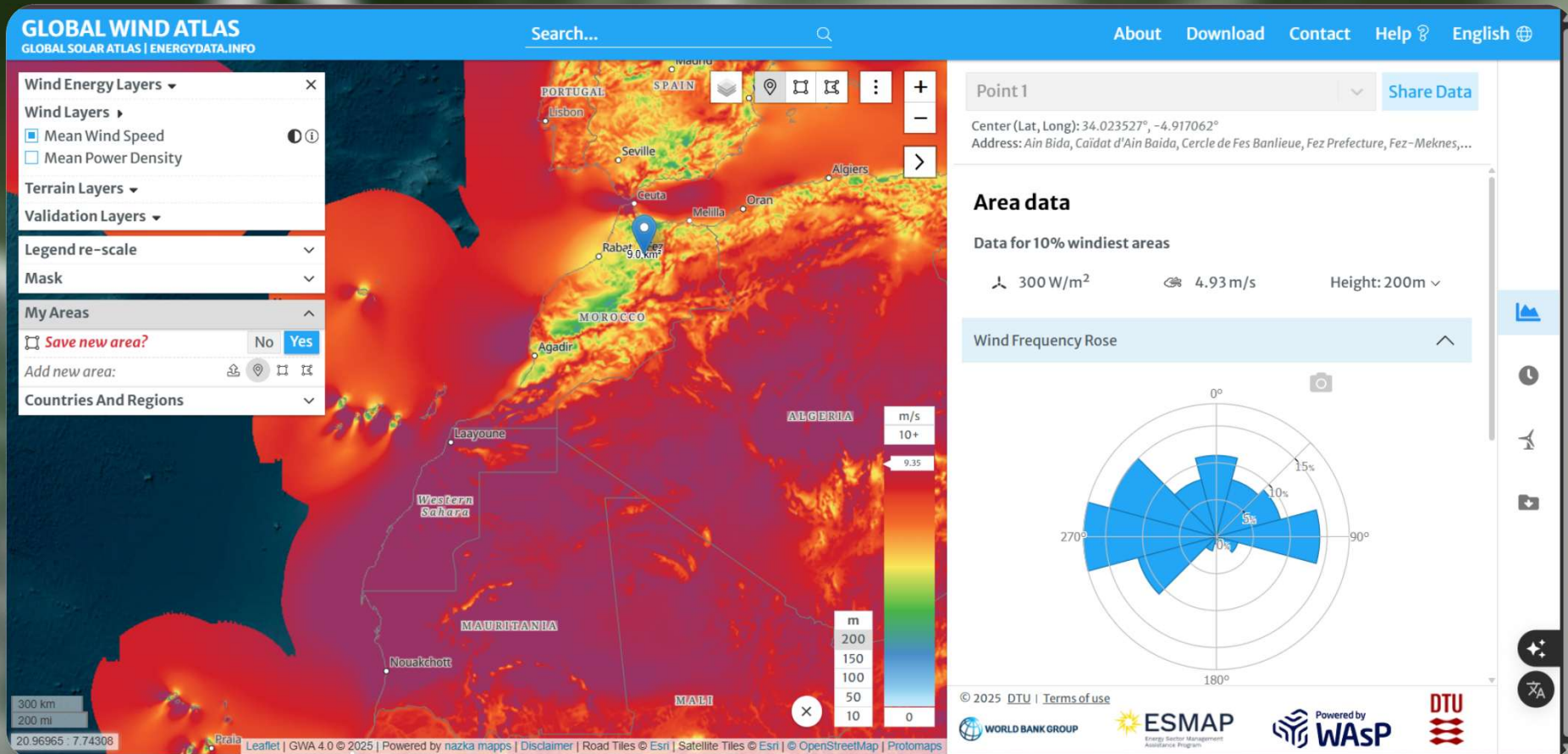


Origine du vent et phénomènes quotidiens

chauffage solaire inégal

Pressure Gradient

Wind (The Result)

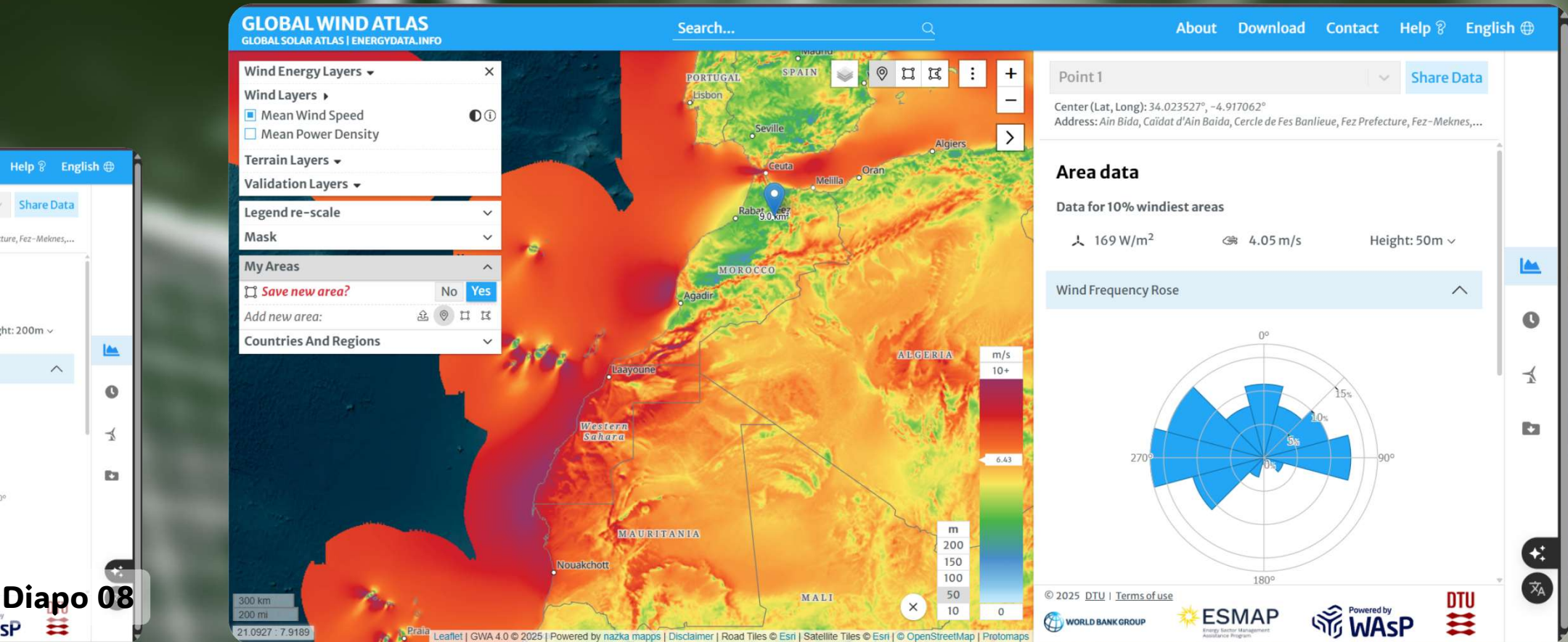


Origine du vent et phénomènes quotidiens

chauffage solaire inégal

Pressure Gradient

Wind (The Result)

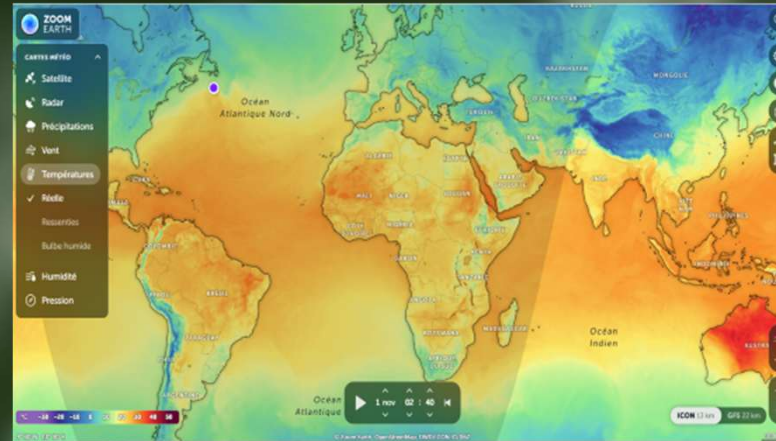


Origine du vent et phénomènes quotidiens

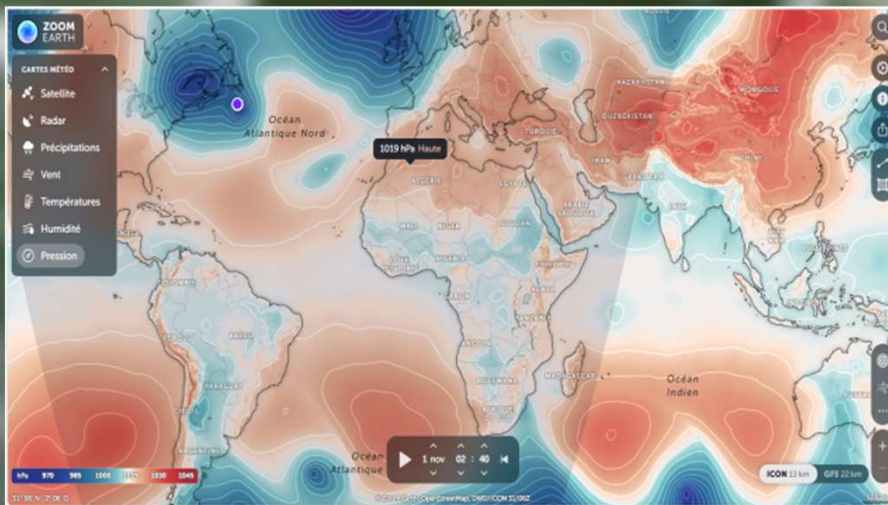
chauffage solaire inégal

Pressure Gradient

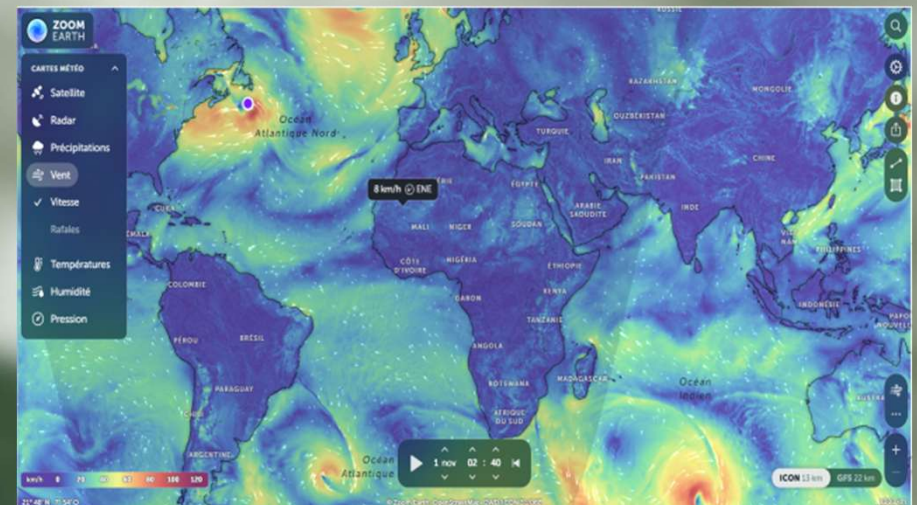
Wind (The Result)



Température



Pression



Vent

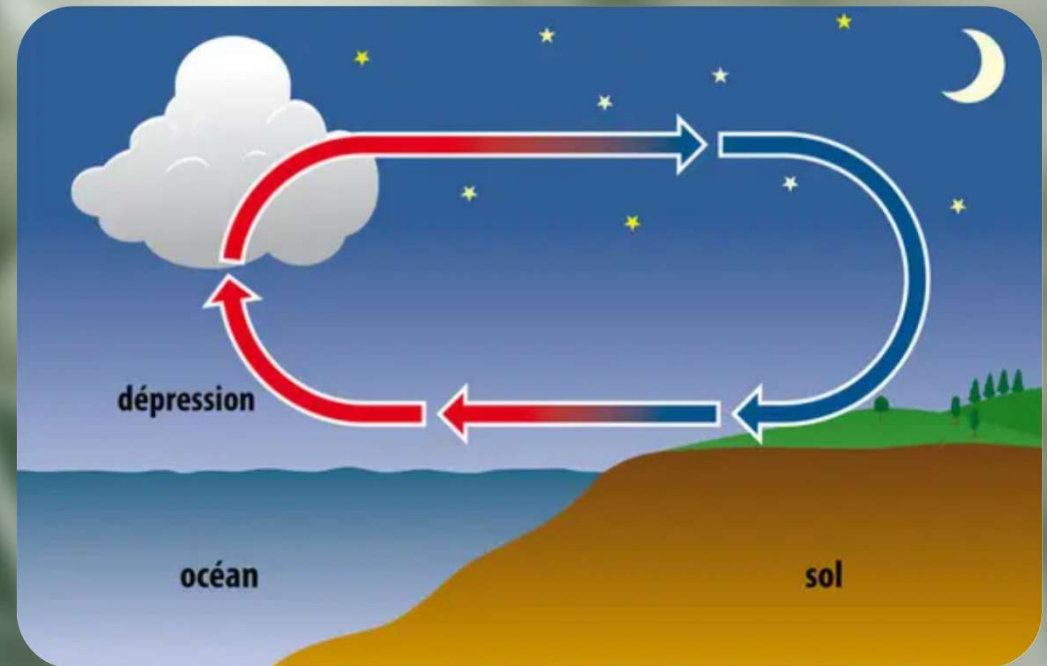
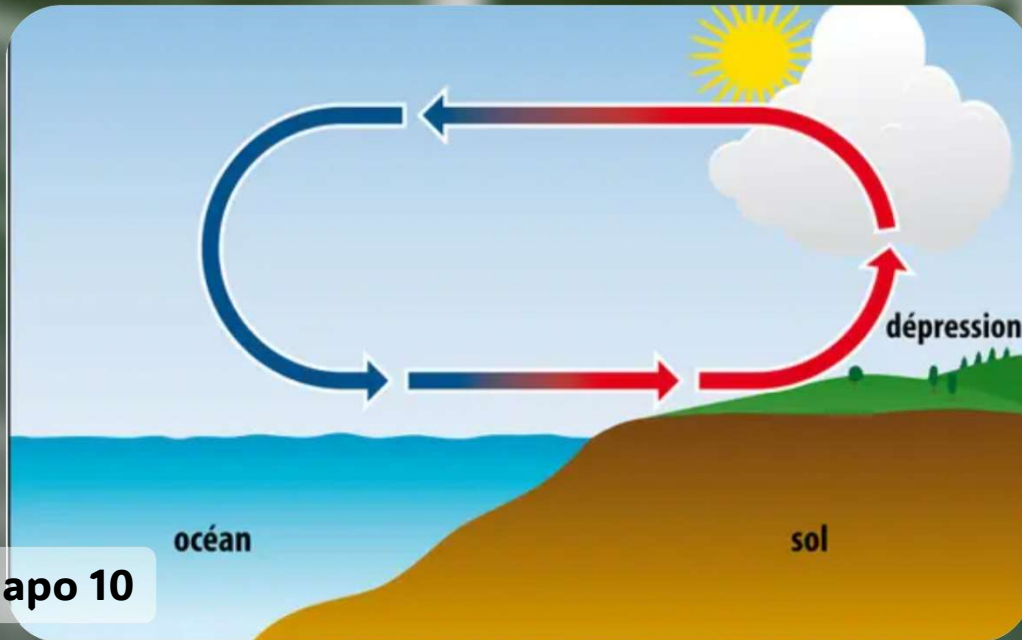
Origine du vent et phénomènes quotidiens

chauffage solaire inégal

Pressure Gradient

Wind (The Result)

Brise de Mer et de Terre



Origine du vent et phénomènes quotidiens

chauffage solaire inégal

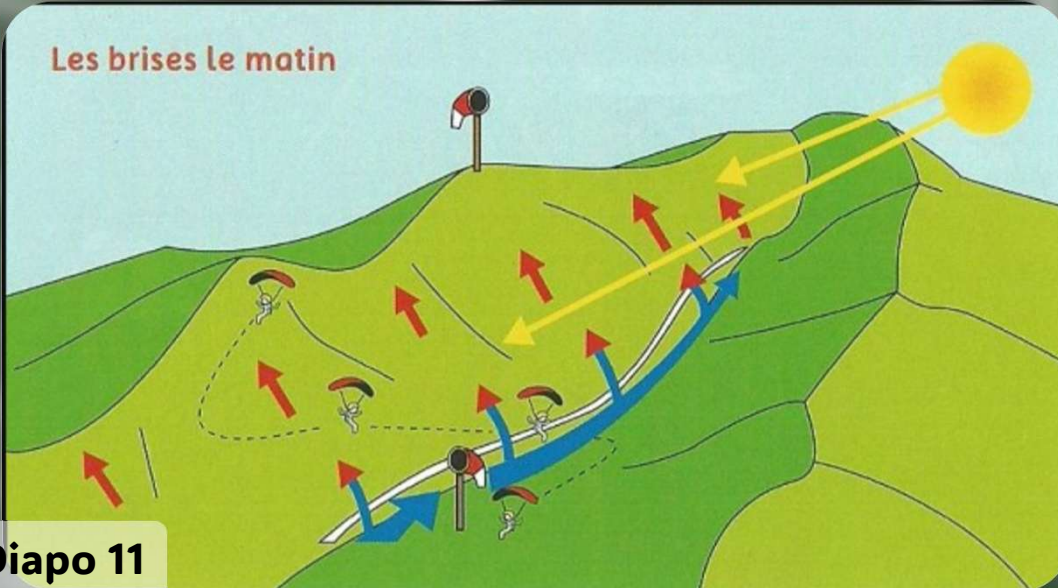
Brise de Mer et de Terre

Pressure Gradient

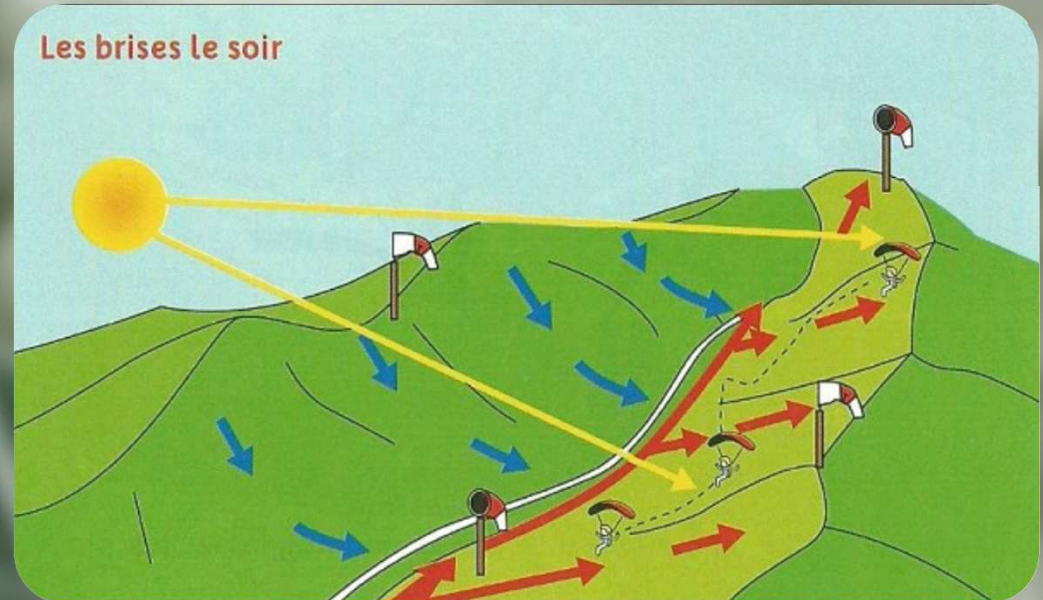
Wind (The Result)

Brise de Montagne

Les brises le matin



Les brises le soir





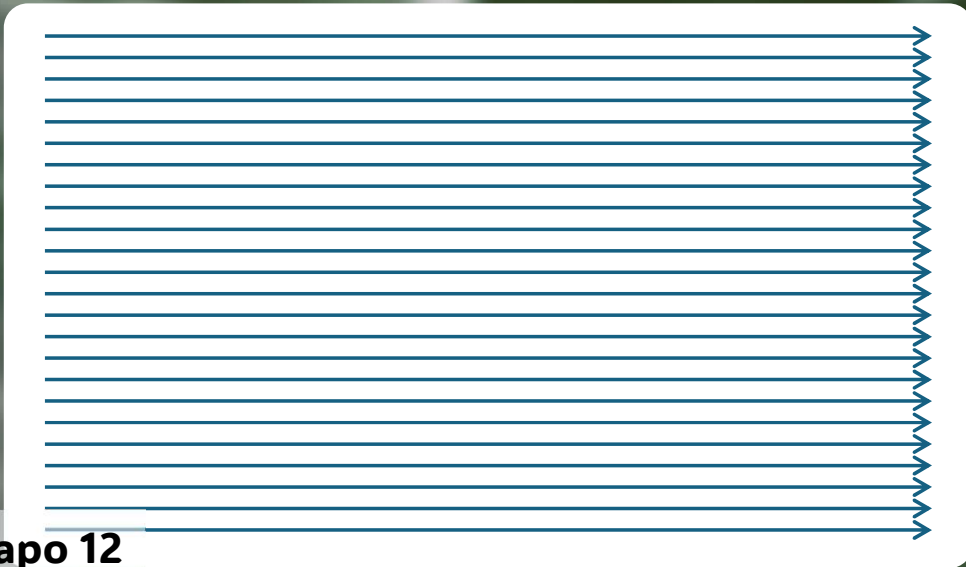
Turbulence et cisaillement

Turbulence et cisaillement

$$R_e = \frac{\rho \cdot V \cdot L}{\mu} = \frac{V \cdot L}{\nu}$$

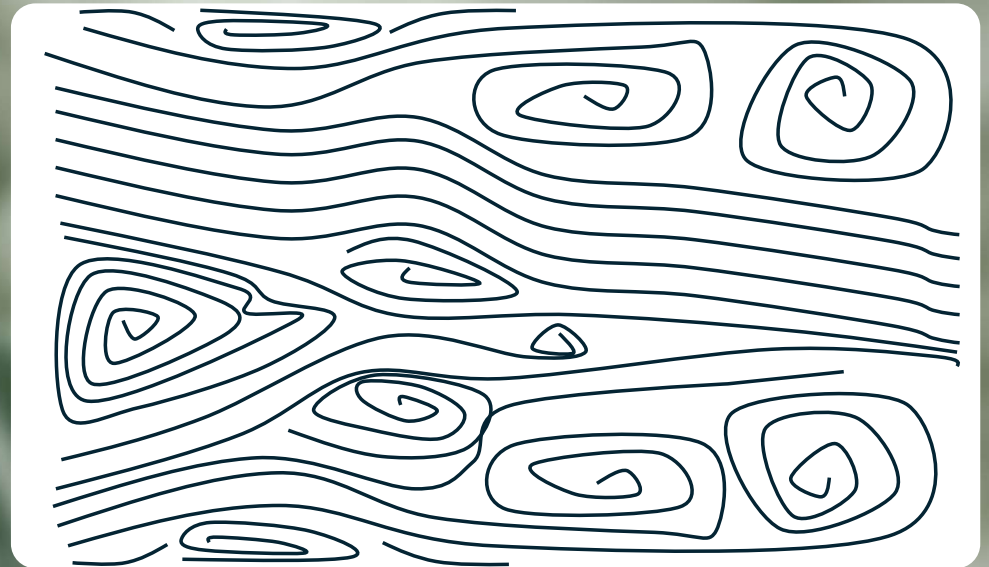
$$R_e < R_{CR}$$

Régime laminaire



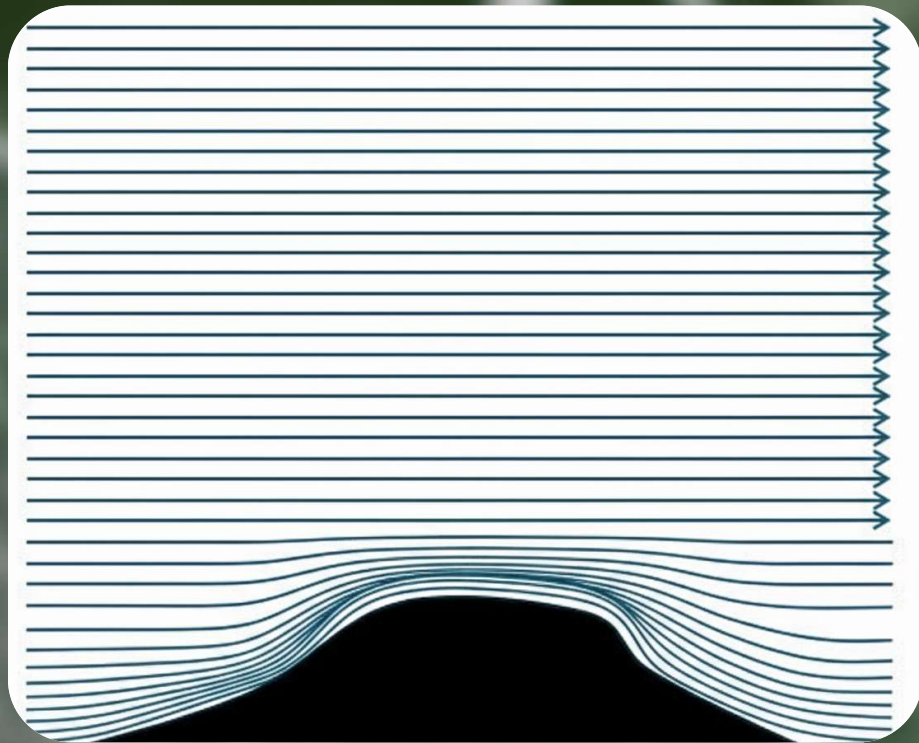
$$R_e > R_{CR} \approx 3000$$

Régime Turbulent



Turbulence et cisaillement

L'impact de la topographie terrestre

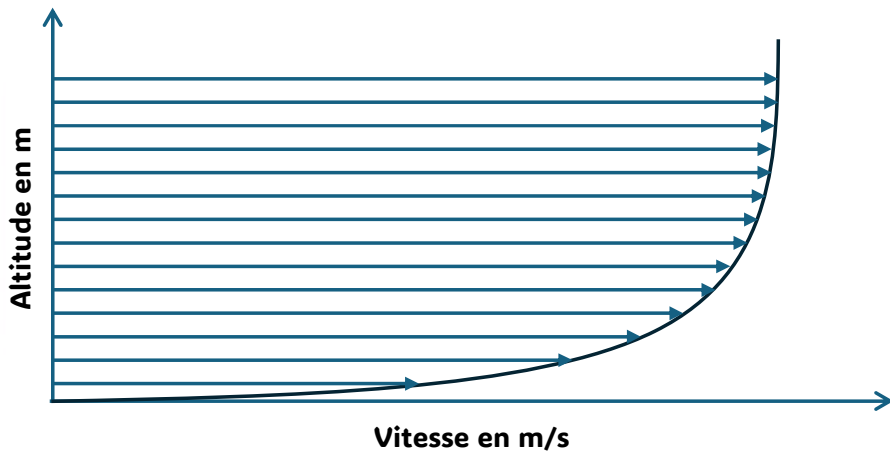


Turbulence et cisaillement

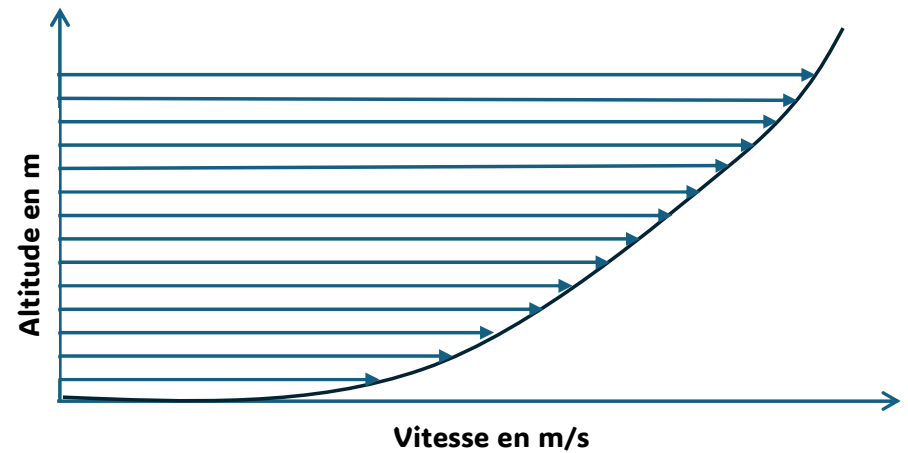
$$\alpha = \frac{\ln \left(\frac{V_2}{V_1} \right)}{\ln \left(\frac{H_2}{H_1} \right)}$$

$$\frac{V_2}{V_1} = \left(\frac{H_2}{H_1} \right)^\alpha$$

Coef=0,1



Coeff=0,6





Principe de fonctionnement

Principe de fonctionnement



Pression

Masse volumique (densité)

Volume

Vitesse

Énergie cinétique $E_c = \frac{1}{2} \cdot m \cdot v^2$

**Couplage
Aérodynamique**



Principe de fonctionnement

Énergie cinétique $E_c = \frac{1}{2} \cdot m \cdot v^2$



Forme aérodynamique

Longueur

Matériaux

Masse

Puissance récupérable $P_{req} = C_p \cdot P_{disponible}$



Couplage
Mecanique



Principe de fonctionnement

Énergie cinétique $E_c = \frac{1}{2} \cdot m \cdot v^2$ Puissance récupérable $P_{req} = C_p \cdot P_{disponible}$

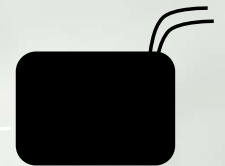


Rapport de transmission

Résistance

Puissance de rotation $P_{rot} = C \cdot \omega$

Couplage
Mecanique



Principe de fonctionnement

Énergie cinétique $E_c = \frac{1}{2} \cdot m \cdot v^2$ Puissance récupérable $P_{req} = C_p \cdot P_{disponible}$ Puissance de rotation $P_{rot} = C \cdot \omega$



Type de générateur utilisé

Puissance nominale P_{Elec}

Rendement η

Tension de sortie V

Puissance de sortie électrique

$$P_{Elec} = \eta \cdot P_{rot}$$

Générateurs Asynchrones

Compacts
rendement élevé

Générateurs synchrones

Robustes
rendement plus faible

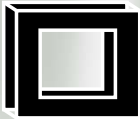
Couplage
Electrique



Principe de fonctionnement

Énergie cinétique $E_c = \frac{1}{2} \cdot m \cdot v^2$ Puissance récupérable $P_{req} = C_p \cdot P_{disponible}$ Puissance de rotation $P_{rot} = C \cdot \omega$

Puissance de sortie électrique $P_{Elec} = \eta \cdot P_{rot}$ Tension de sortie V

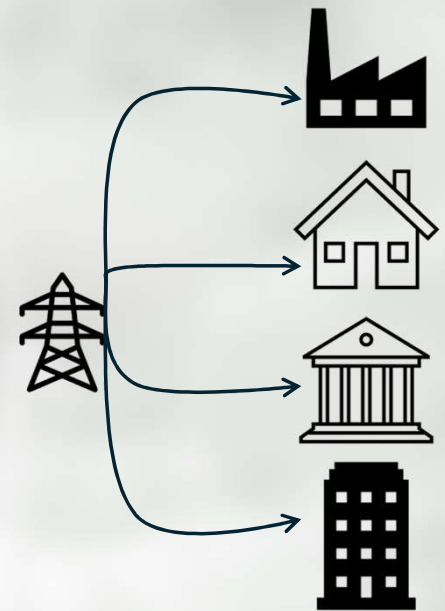


Puissance nominale

Niveau de tension (entré et sortie)

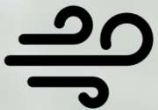
Rendement > 90%

Refroidissement



Principe de fonctionnement

Énergie cinétique $E_c = \frac{1}{2} \cdot m \cdot v^2$ Puissance récupérable $P_{req} = C_p \cdot P_{disponible}$ Puissance de rotation $P_{rot} = C \cdot \omega$
Puissance de sortie électrique $P_{Elec} = \eta \cdot P_{rot}$ Tension de sortie V



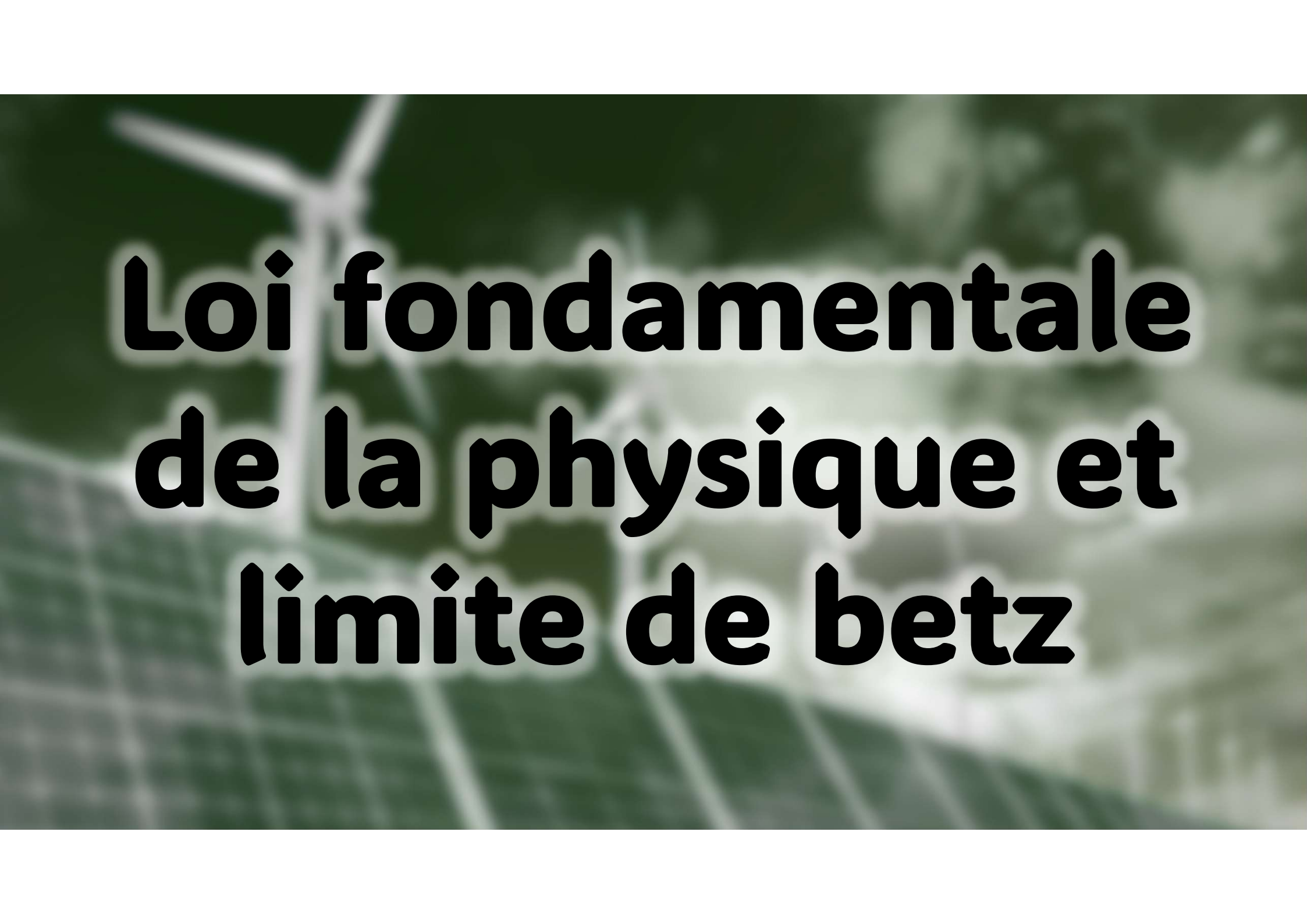
Énergie
cinétique



Énergie mécanique



Énergie électrique

The background of the slide is a blurred image of a wind turbine, showing its three blades and central hub against a soft-focus landscape.

Loi fondamentale de la physique et limite de betz

Loi fondamentale de la physique

Conservation de Masse

$$\dot{m} = \rho \cdot S \cdot v$$

Quantité de mouvement

$$F_{D/f} = \dot{m} \cdot (v_0 - v)$$

$$F_{D/f} = S \cdot (P - P')$$

Equation de Bernoulli (sans machines)

$$P_1 + \frac{1}{2} \cdot \rho \cdot (v_1)^2 + g z_1 = P_2 + \frac{1}{2} \cdot \rho \cdot (v_2)^2 + g z_2$$

Puissance (en fonction de quantité de mouvement)

$$p_1 = F_{D/f} \cdot v$$

Puissance cinétique

$$P_c = \frac{1}{2} \cdot \dot{m} \cdot v^2$$

Puissance récupérable

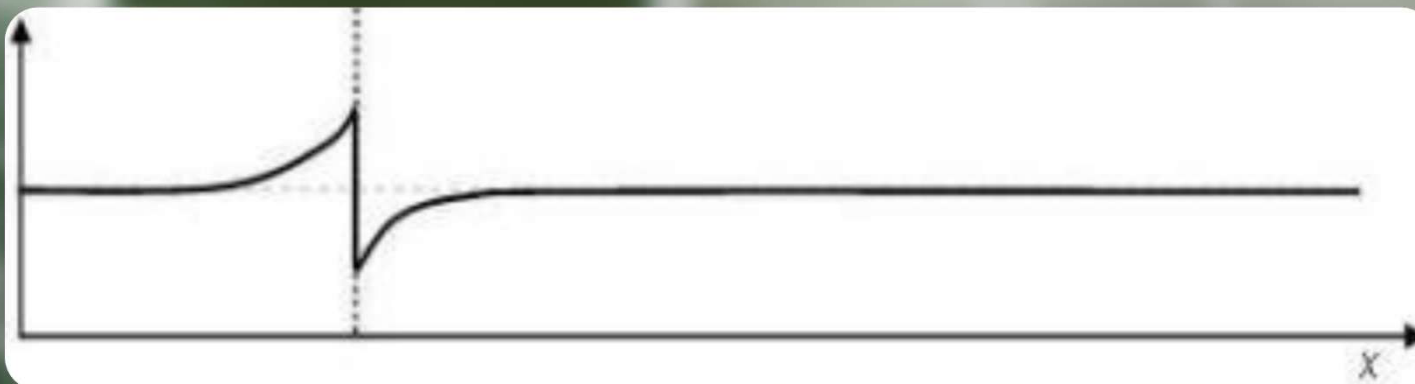
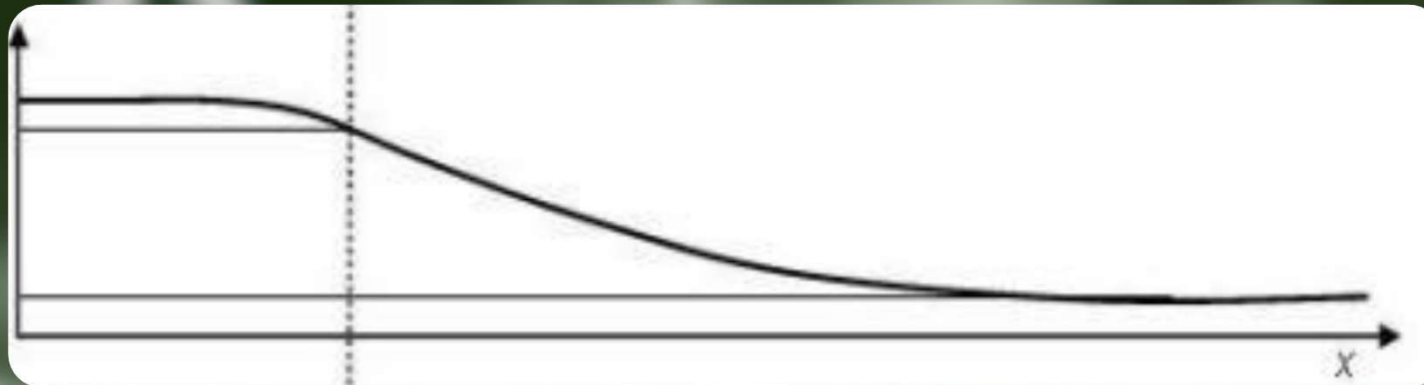
$$P_{req} = C_p \cdot P_{disponible}$$

Facteur d'interférence

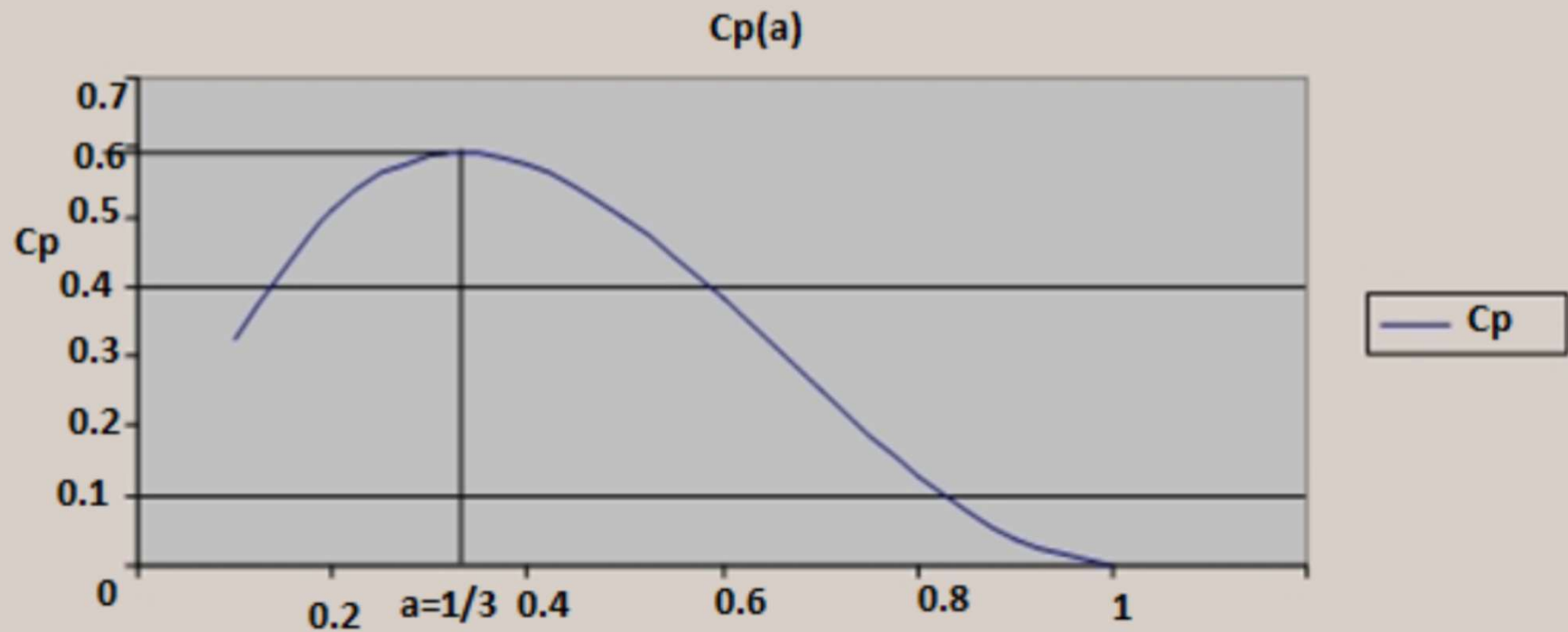
$$v = (1 - \alpha) \cdot v_0$$

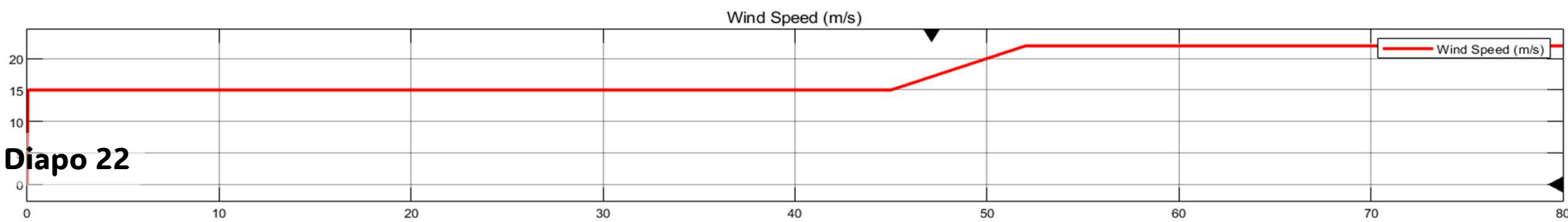
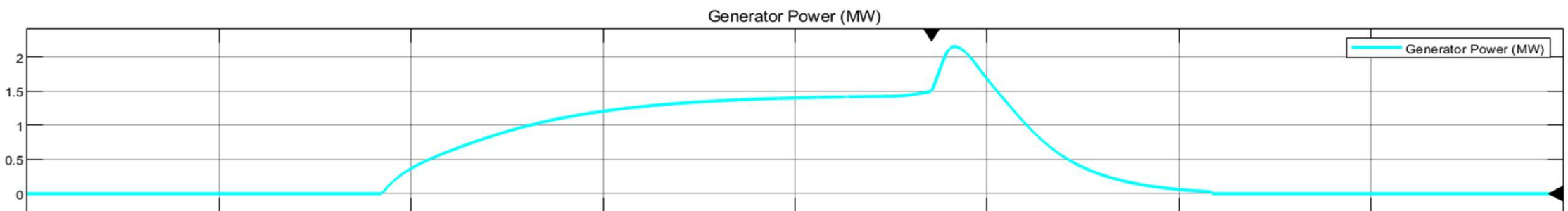
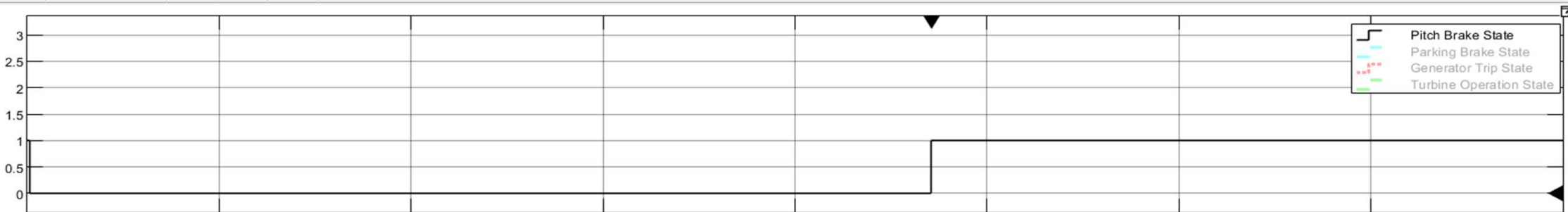
$$v_1 = (1 - 2\alpha) \cdot v_0$$

Limite de Betz



Limite de Betz





Diapo 22



Atlas de vent



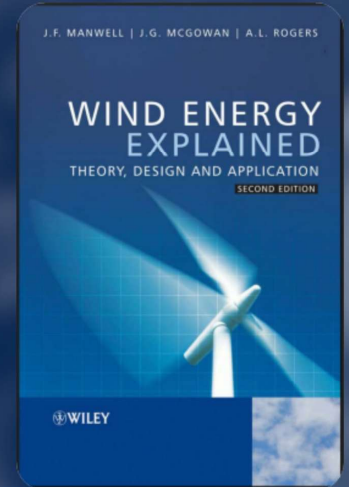
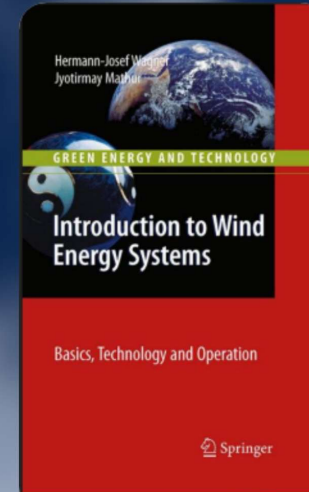
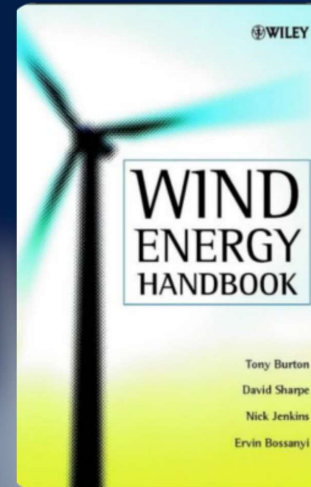
**Aérodynamique d'une
éolienne à axe
horizontal**



Gradient de pression



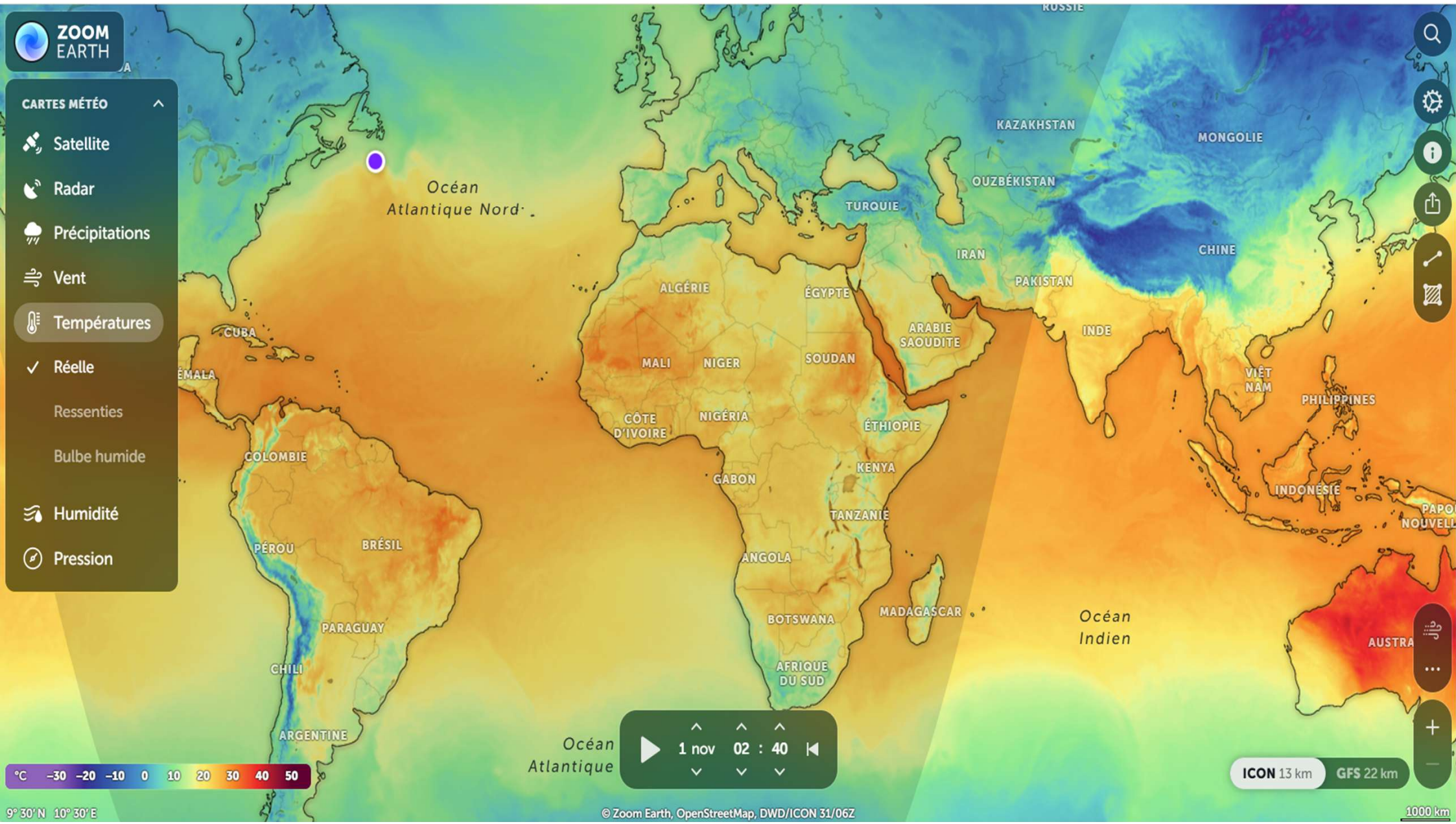
**3D
wind-energy**



https://re.jrc.ec.europa.eu/pvg_tools/en/

<https://www.nrel.gov/research/learning>

<https://sam.nrel.gov/download.html>





CARTES MÉTÉO

- Satellite
- Radar
- Précipitations
- Vent
- Températures
- Humidité
- Pression

hPa 970 985 1000 1015 1030 1045

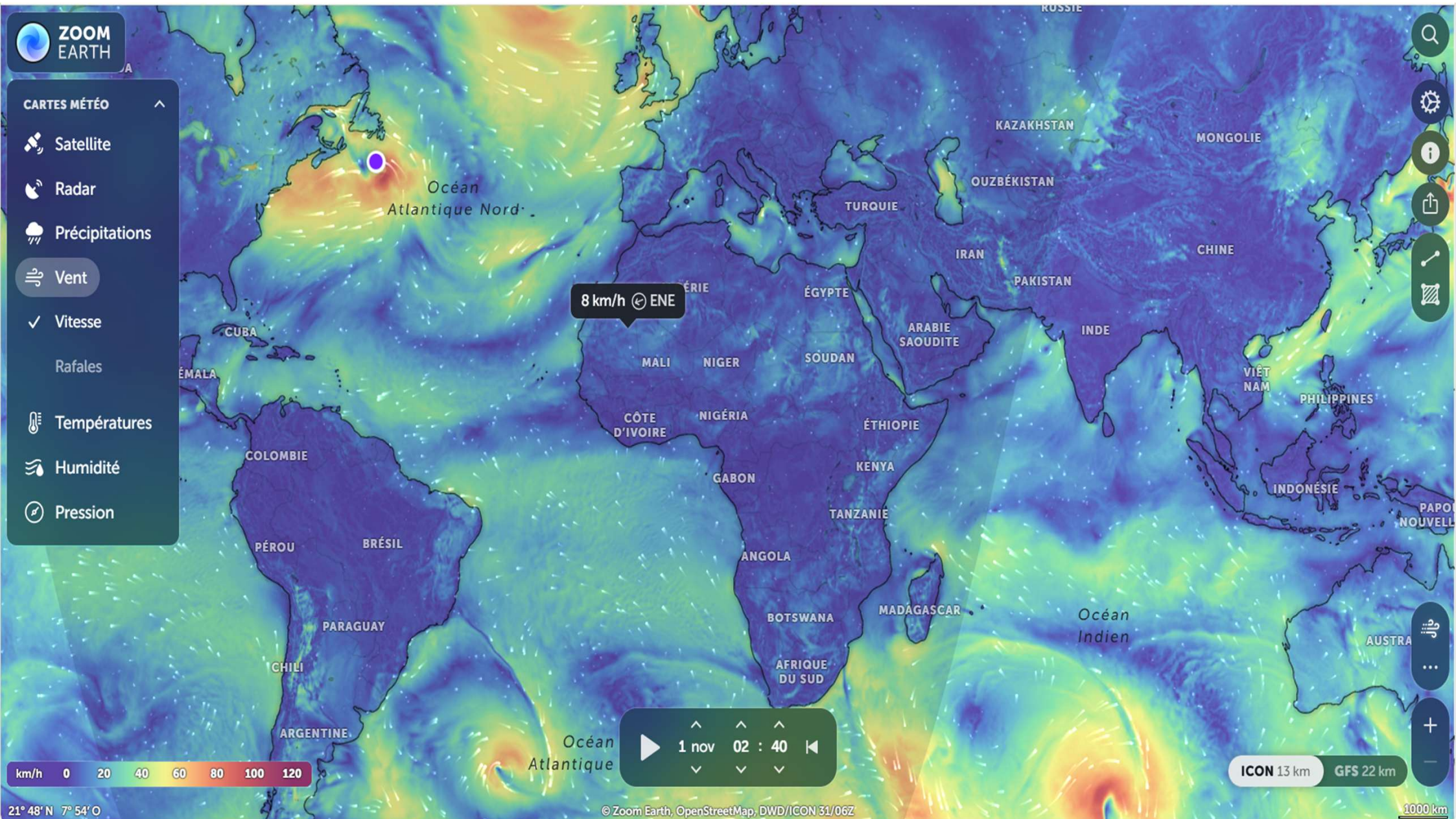
1019 hPa Haute

1 nov 02 : 40

ICON 13 km GFS 22 km

© Zoom Earth, OpenStreetMap, DWD/ICON 31/06Z

1000 km



Loi fondamentale de la physique

Conservation de Masse

$$\dot{m} = \rho \cdot S \cdot v$$

Quantité de mouvement

$$F_{D/f} = \dot{m} \cdot (v_0 - v)$$

$$F_{D/f} = S \cdot (P - P')$$

Equation de Bernoulli (sans machines)

$$P_1 + \frac{1}{2} \cdot \rho \cdot (v_1)^2 + g z_1 = P_2 + \frac{1}{2} \cdot \rho \cdot (v_2)^2 + g z_2$$

Puissance (en fonction de quantité de mouvement)

$$p_1 = F_{D/f} \cdot v$$

Puissance cinétique

$$P_c = \frac{1}{2} \cdot \dot{m} \cdot v^2$$

Puissance récupérable

$$P_{req} = C_p \cdot P_{disponible}$$

Facteur d'interférence

$$v = (1 - \alpha) \cdot v_0$$

$$v_1 = (1 - 2\alpha) \cdot v_0$$

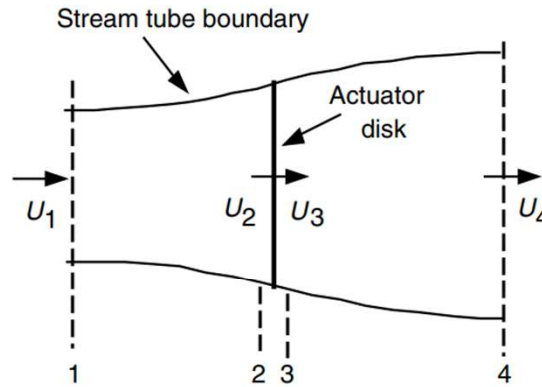


Figure 3.1 Actuator disc model of a wind turbine; U , mean air velocity; 1, 2, 3, and 4 indicate locations

Applying the conservation of linear momentum to the control volume enclosing the whole system, one can find the net force on the contents of the control volume. That force is equal and opposite to the thrust, T , which is the force of the wind on the wind turbine. From the conservation of linear momentum for a one-dimensional, incompressible, time-invariant flow, the thrust is equal and opposite to the rate of change of momentum of the air stream:

$$T = U_1(\rho AU)_1 - U_4(\rho AU)_4 \quad (3.1)$$

where ρ is the air density, A is the cross-sectional area, U is the air velocity, and the subscripts indicate values at numbered cross-sections in Figure 3.1.

For steady state flow, $(\rho AU)_1 = (\rho AU)_4 = \dot{m}$, where $\dot{m}$ is the mass flow rate. Therefore:

$$T = \dot{m}(U_1 - U_4) \quad (3.2)$$

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$$T = \dot{m}(U_1 - U_4) \quad (3.2)$$

The thrust is positive so the velocity behind the rotor, U_4 , is less than the free stream velocity, U_1 . No work is done on either side of the turbine rotor. Thus the Bernoulli function can be used in the two control volumes on either side of the actuator disc. In the stream tube upstream of the disc:

$$p_1 + \frac{1}{2}\rho U_1^2 = p_2 + \frac{1}{2}\rho U_2^2 \quad (3.3)$$

In the stream tube downstream of the disc:

$$p_3 + \frac{1}{2}\rho U_3^2 = p_4 + \frac{1}{2}\rho U_4^2 \quad (3.4)$$

where it is assumed that the far upstream and far downstream pressures are equal ($p_1 = p_4$) and that the velocity across the disc remains the same ($U_2 = U_3$).

$$p_3 + \frac{1}{2}\rho U_3^2 = p_4 + \frac{1}{2}\rho U_4^2 \quad (3.4)$$

where it is assumed that the far upstream and far downstream pressures are equal ($p_1 = p_4$) and that the velocity across the disc remains the same ($U_2 = U_3$).

The thrust can also be expressed as the net sum of the forces on each side of the actuator disc:

$$T = A_2(p_2 - p_3) \quad (3.5)$$

If one solves for $(p_2 - p_3)$ using Equations (3.3) and (3.4) and substitutes that into Equation (3.5), one obtains:

$$T = \frac{1}{2}\rho A_2(U_1^2 - U_4^2) \quad (3.6)$$

Equating the thrust values from Equations (3.2) and (3.6) and recognizing that the mass flow rate is also $\rho A_2 U_2$, one obtains:

$$U_2 = \frac{U_1 + U_4}{2} \quad (3.7)$$

Thus, the wind velocity at the rotor plane, using this simple model, is the average of the upstream and downstream wind speeds.

If one defines the axial induction factor, a , as the fractional decrease in wind velocity between the free stream and the rotor plane, then

$$a = \frac{U_1 - U_2}{U_1} \quad (3.8)$$

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$$a = \frac{U_1 - U_2}{U_1} \quad (3.8)$$

$$U_4 = U_1(1 - 2a) \quad (3.10)$$

The quantity $U_1 a$ is often referred to as the induced velocity at the rotor, in which case the velocity of the wind at the rotor is a combination of the free stream velocity and the induced wind velocity. As the axial induction factor increases from 0, the wind speed behind the rotor slows more and more. If $a = 1/2$, the wind has slowed to zero velocity behind the rotor and the simple theory is no longer applicable.

The power out, P , is equal to the thrust times the velocity at the disc:

$$P = \frac{1}{2} \rho A_2 (U_1^2 - U_4^2) U_2 = \frac{1}{2} \rho A_2 U_2 (U_1 + U_4) (U_1 - U_4) \quad (3.11)$$

Substituting for U_2 and U_4 from Equations (3.9) and (3.10) gives:

$$P = \frac{1}{2} \rho A U^3 4a(1 - a)^2 \quad (3.12)$$

where the control volume area at the rotor, A_2 , is replaced by A , the rotor area, and the free stream velocity U_1 is replaced by U .

Wind turbine rotor performance is usually characterized by its power coefficient, C_P :

$$C_P = \frac{P}{\frac{1}{2} \rho U^3 A} = \frac{\text{Rotor power}}{\text{Power in the wind}} \quad (3.13)$$

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$$C_P = \frac{P}{\frac{1}{2}\rho U^3 A} = \frac{\text{Rotor power}}{\text{Power in the wind}} \quad (3.13)$$

$$C_P = 4a(1 - a)^2 \quad (3.14)$$

The maximum C_P is determined by taking the derivative of the power coefficient (Equation (3.14)) with respect to a and setting it equal to zero, yielding $a = 1/3$. Thus:

$$C_{P,\max} = 16/27 = 0.5926 \quad (3.15)$$

when $a = 1/3$. For this case, the flow through the disc corresponds to a stream tube with an upstream cross-sectional area of $2/3$ the disc area that expands to twice the disc area downstream. This result indicates that, if an ideal rotor were designed and operated such that the wind speed at the rotor were $2/3$ of the free stream wind speed, then it would be operating at the point of maximum power production. Furthermore, given the basic laws of physics, this is the maximum power possible.

From Equations (3.6), (3.9) and (3.10), the axial thrust on the disc is:

$$T = \frac{1}{2}\rho A U^2 [4a(1 - a)] \quad (3.16)$$

Similarly to the power, the thrust on a wind turbine can be characterized by a non-dimensional thrust coefficient:

$$C_T = \frac{T}{\frac{1}{2}\rho U^2 A} = \frac{\text{Thrust force}}{\text{Dynamic force}} \quad (3.17)$$

thrust coefficient:

$$C_T = \frac{T}{\frac{1}{2}\rho U^2 A} = \frac{\text{Thrust force}}{\text{Dynamic force}} \quad (3.17)$$

